

365 Data Science SQL Basics – Topic 1.2: Data Manipulation Language (DML)

Exhaustive Notes on DML Statements: INSERT, UPDATE, DELETE & Row Manipulations

1. Data Manipulation Language (DML) Overview

DML commands manage data records inside existing table structures. Unlike DDL, DML directly affects table rows without modifying schema definitions.

2. Core DML Commands Breakdown

- `INSERT`: Adds new record rows into tables (`INSERT INTO table VALUES (...)`).
- `UPDATE`: Modifies attribute values of existing records (`UPDATE table SET col = val WHERE ...`).
- `DELETE`: Removes specific record rows conditionally (`DELETE FROM table WHERE ...`).

```
-- DML Operations Example
INSERT INTO customers (first_name, last_name) VALUES ('Asha', 'Sharma');
UPDATE customers SET last_name = 'Johnson' WHERE customer_id = 1;
DELETE FROM customers WHERE customer_id = 10;
```

3. Topic 1.2 Viva Q&A

1. Q: What happens if an UPDATE statement is executed without a WHERE clause? A: All rows in the table will be updated to the specified new value!